

Unsupervised Learning - Lab 11 - Bayesian Networks

Structure Learning with and without Expert's Knowledge

Alessio Zanga^{1, 2, *}

May 12, 2025

¹Models and Algorithms for Data and Text Mining (MADLab), University of Milano-Bicocca, Milan, Italy

²School of Medicine and Surgery, University of Milano-Bicocca, Monza, Italy

*Corresponding author: alessio.zanga@unimib.it

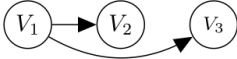
- Structure learning is the task of identifying the optimal Directed Acyclic Graph (DAG) that captures the dependencies among variables.
- Two main approaches:
 - **Constraint-based:** Uses statistical tests to infer independence relationships.
 - **Score-based:** Searches for the best structure according to a scoring function.
- Hybrid methods combine both approaches to leverage their strengths.

- Scoring functions evaluate how well a structure fits the data.
- Common scoring criteria:
 - **BIC (Bayesian Information Criterion)**: Penalizes model complexity.
 - **AIC (Akaike Information Criterion)**: Balances goodness of fit.
 - **Bayesian Dirichlet (BD) scores**: Use priors and likelihood.
- The higher the score, the better the structure (depending on criterion).

Hill-Climbing Algorithm

- A greedy search algorithm used in score-based structure learning.
 1. Starts with an initial network (often empty or naive Bayes).
 2. Iteratively applies local changes (add, remove, reverse an edge).
 3. Compute the score improvement.
 4. Stops when no further improvement is possible.
- Many local optima, restarts or tabu search can mitigate the problem.

Hill-Climbing Algorithm

current DAG state $\mathcal{G}$	edge operation	neighbouring DAG $\mathcal{G}_{nei}$
	add $V_1 \rightarrow V_3$	
	add $V_2 \rightarrow V_3$	
	add $V_3 \rightarrow V_1$	
	add $V_3 \rightarrow V_2$	
	reverse $V_1 \rightarrow V_2$	
	delete $V_1 \rightarrow V_2$	

Expert's Knowledge - Forbidden and Required Edges

- Domain experts can specify constraints:
 - **Forbidden edges:** Edges that must not appear in the network.
 - **Required edges:** Edges that must be included.
- These constraints guide the learning process, improve interpretability, and reduce the search space.
- Especially useful in domains with strong prior knowledge (e.g., medicine, ...).

- In many domains, the temporal order of variables is known (cause precedes effect).
- Temporal constraints can enforce a **partial ordering over nodes**.
- Prevents cycles and improves the plausibility of learned structures.
- Common in time-series, decision support systems, and diagnosis models.

- Bayesian Networks model complex probabilistic relationships using DAGs.
- Structure learning identifies the optimal network from data and prior knowledge.
- Scoring functions and search algorithms (like hill-climbing) are key tools.
- Expert knowledge significantly enhances model quality and learning efficiency.